

DNS Recon

CompTIA PenTest+ (PT0-003)

Domain:

- 2.0: Reconnaissance and Enumeration

Objectives:

- 2.1: Given a scenario, apply information gathering techniques
- 2.2: Given a scenario, apply enumeration techniques.

Video Overview

This video provides an overview of DNS reconnaissance as a fundamental information-gathering technique in penetration testing. You will learn how to verify target ownership, identify associated hosts, and uncover network infrastructure details using various DNS tools. Understanding these techniques is applicable not only in cybersecurity roles but also in broader IT fields for network troubleshooting, infrastructure mapping, and enhancing organizational security posture. This knowledge will help you grasp how external attackers view and approach target systems, preparing you for real-world scenarios in information security.

Key Topics Covered

- **DNS reconnaissance as an initial step in information gathering for penetration testing:** Understanding the importance of DNS information for verifying system ownership and identifying in-scope targets.
- **Using NSlookup for forward DNS resolution (domain to IP address):** How to translate domain names into their corresponding IP addresses, noting when multiple IPs indicate load balancing or CDN use.
- **Alternative DNS tools: dig and host (for Linux systems):** Exploring additional tools beyond NSlookup that offer similar DNS resolution capabilities with different output formats.
- **Utilizing the host command to reveal mail services and other infrastructure details:** How to gain insights into a target's infrastructure, such as email providers, which can inform potential social engineering attacks.

- **Identifying primary and secondary DNS servers using host -t ns:** Learning to query for name servers to understand an organization's DNS hosting, including the use of cloud services like AWS.
- **Performing reverse DNS lookups (IP address to domain) with dig -x:** Techniques for resolving an IP address back to its associated domain name, helping to map networks and uncover unexpected connections.
- **Understanding DNS zone transfers as a vulnerability and their use in enumeration:** Recognizing that misconfigured zone transfers can expose complete databases of internal network details to attackers.

Student Notes Section